

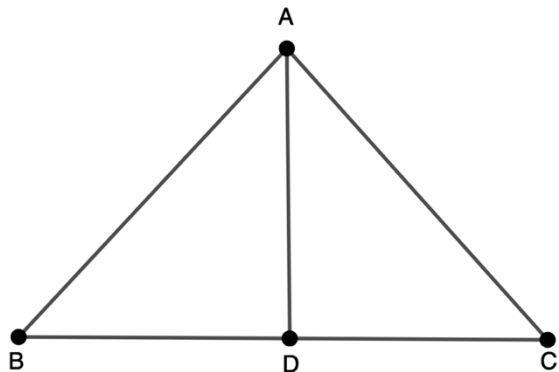
Geometry
Unit 2
Lesson 8 Practice

Name _____
Date _____
Period _____

Complete the two-column proof.

Given: $\angle ADB$ is a right angle, $\angle ADB$ and $\angle ADC$ are supplementary.

Prove: $\angle ADB \cong \angle ADC$

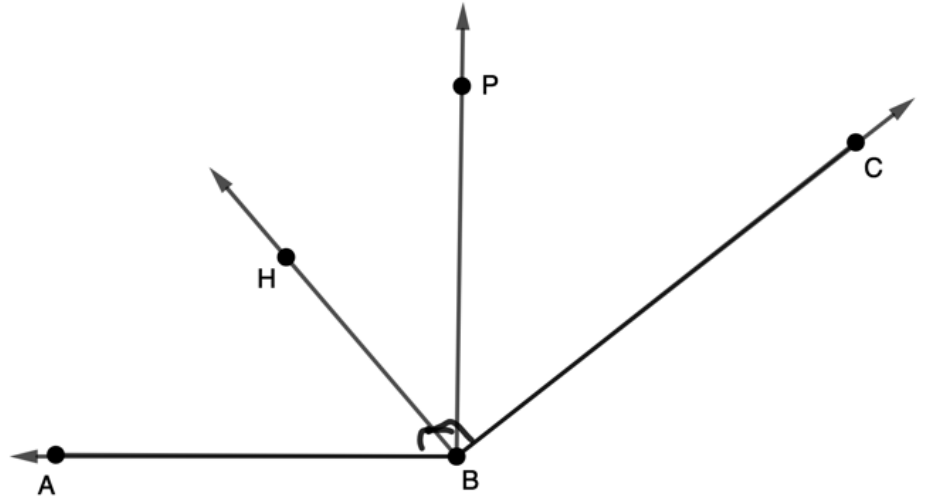


	Statement	Reason
1.	$\angle ADB$ is a right angle	Given
2.		Definition of right angle
3.	$\angle ADB$ and $\angle ADC$ are supplementary	Given
4.	$m\angle ADC + m\angle ADB = 180^\circ$	
5.		Substitution Property
6.	$m\angle ADC = 90^\circ$	Subtraction Property
7.		Transitive Property of Equality
8.	$\angle ADC \cong \angle ADB$	

Complete the paragraph proof to answer questions 5-10.

Given: $\angle PBA$ and $\angle HBC$ are right angles

Prove: $\angle ABH \cong \angle PBC$



It is given that $\angle PBA$ and $\angle HBC$ are right angles. Using the addition postulate,

6. _____ and 7. _____.

Using the 8. _____ property of equality, we know

$m\angle ABH + m\angle HBP = m\angle HBP + m\angle PBC$. Next, we apply the subtraction property of equality

to determine 9. _____. By the definition of 10. _____,

$\angle ABH \cong \angle PBC$.